Project : Implementing and Managing Active Directory in Windows Server

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1. Project Objective

To set up and manage Active Directory in a Windows Server environment, including configuring domain controllers, structuring organizational units, managing user accounts, and implementing security policies.

2. Project Description

This project highlights the value of centralized identity management, security, and resource allocation. Implementing Active Directory enhances authentication, enforces security policies, and simplifies user and group management for better cybersecurity and operational efficiency.

3. Real-Time Scenario Context

Organizations often struggle with managing user authentication, resource access, and security policies in large-scale IT environments. Without a structured Active Directory (AD) implementation, companies face challenges such as unauthorized access, inefficient user management, and weak policy enforcement, leading to security vulnerabilities and operational inefficiencies.

This project focuses on deploying and managing Active Directory in a Windows server environment to centralize authentication, define organizational units, and enforce security policies through Group Policy Objects (GPOs). By structuring the AD environment, the project enhances network security, streamlines administrative tasks, and ensures compliance with enterprise security standards. The goal is to improve access control, reduce security risks, and optimize IT management in an enterprise setting.

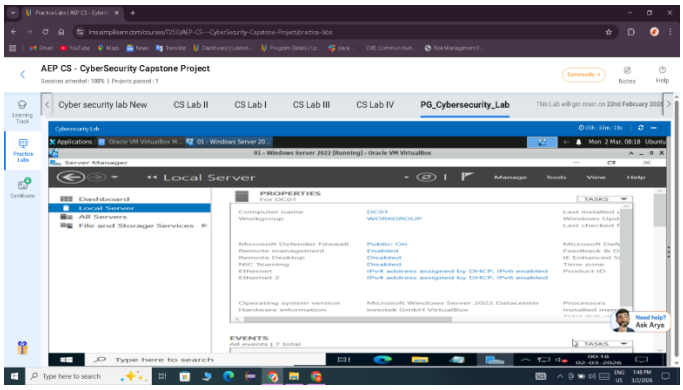
4. Industry Tools Utilized

1. **Windows Servers:** A widely used OS for managing Active Directory and enterprise networks.
2. **Active Directory Domain Services (AD DS):** Enables centralized user authentication and policy enforcement.
3. **Group Policy Objects (GPOs):** Used to enforce security policies and system configurations.
4. **Windows Defender Firewall:** Protects networks by enforcing security rules and monitoring traffic.

5. Execution Methodology & Configuration Log

Phase 1: Server Preparation & Static Identity Assignment

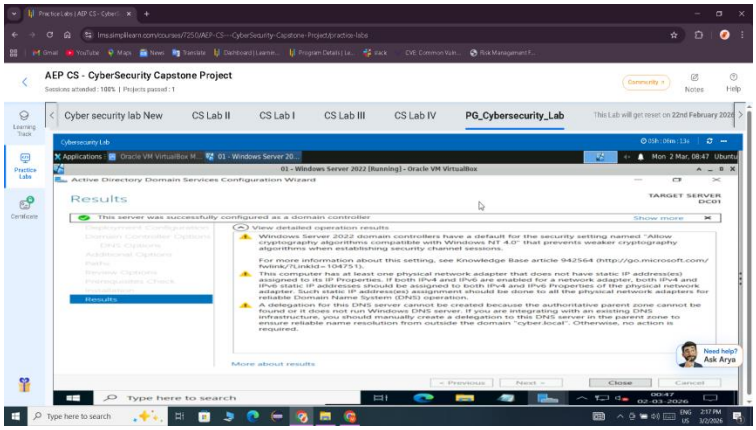
Description: A Domain Controller must have a stable, recognizable identity on the network before Active Directory can be installed. In this phase, the Windows Server 2022 machine was accessed via Server Manager, and its default randomized hostname was permanently changed to `DC01`.



Windows Server Manager Local Server dashboard confirming the computer name has been successfully updated to DC01.

Phase 2: Installing Active Directory Domain Services (AD DS)

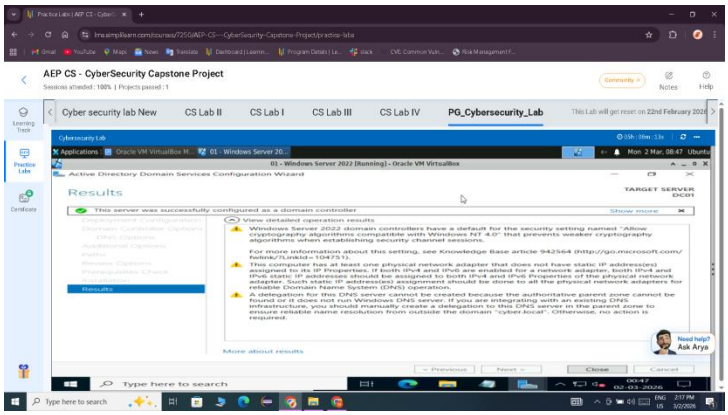
Description: To prepare the server for centralized directory management, the Active Directory Domain Services (AD DS) role was installed. The Add Roles and Features Wizard was utilized to deploy the core AD binaries, Active Directory Administrative Center, and necessary command-line management tools.



Installation progress window confirming the successful installation of the AD DS role, triggering the prompt to promote the server.

Phase 3: Promoting the Server to a Domain Controller

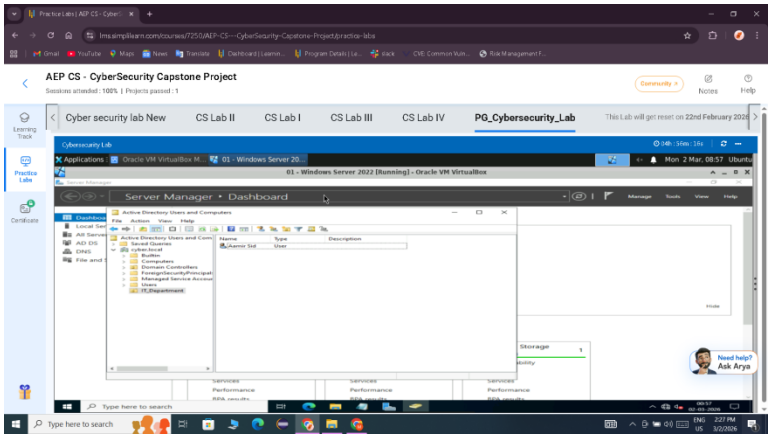
Description: With the foundation laid, the server was officially promoted to a Domain Controller. A new organizational forest was created, establishing the root domain name as `cyber.local`. The system passed all prerequisite checks, allowing the directory partition to be securely built and initialized.



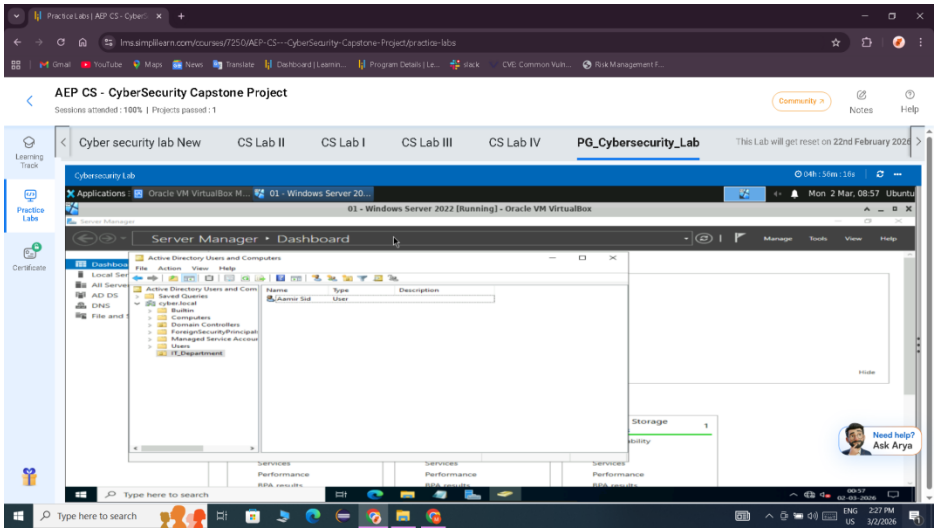
Configuration wizard displaying the green prerequisite checkmark and confirming the server was successfully configured as a domain controller.

Phase 4: Structuring Organizational Units (OUs) & User Accounts

Description: To align the directory with a real-world enterprise structure, the Active Directory Users and Computers console was utilized. A dedicated Organizational Unit (OU) named `IT_Department` was created to group specific personnel. Subsequently, a new employee user account (Aamir Sid / Logon: `Aami`) was successfully provisioned within this OU.



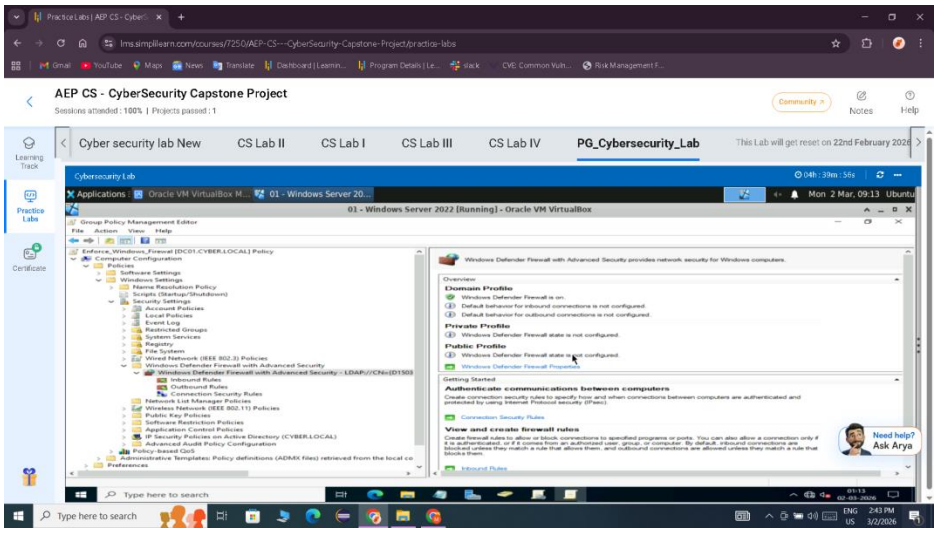
Active Directory console displaying the newly established cyber.local domain tree and default containers.



Verification of the newly provisioned 'Aamir Sid' user account residing securely within the designated IT_Department OU.

Phase 5: Implementing Security Policies (GPOs) & Firewall Rules

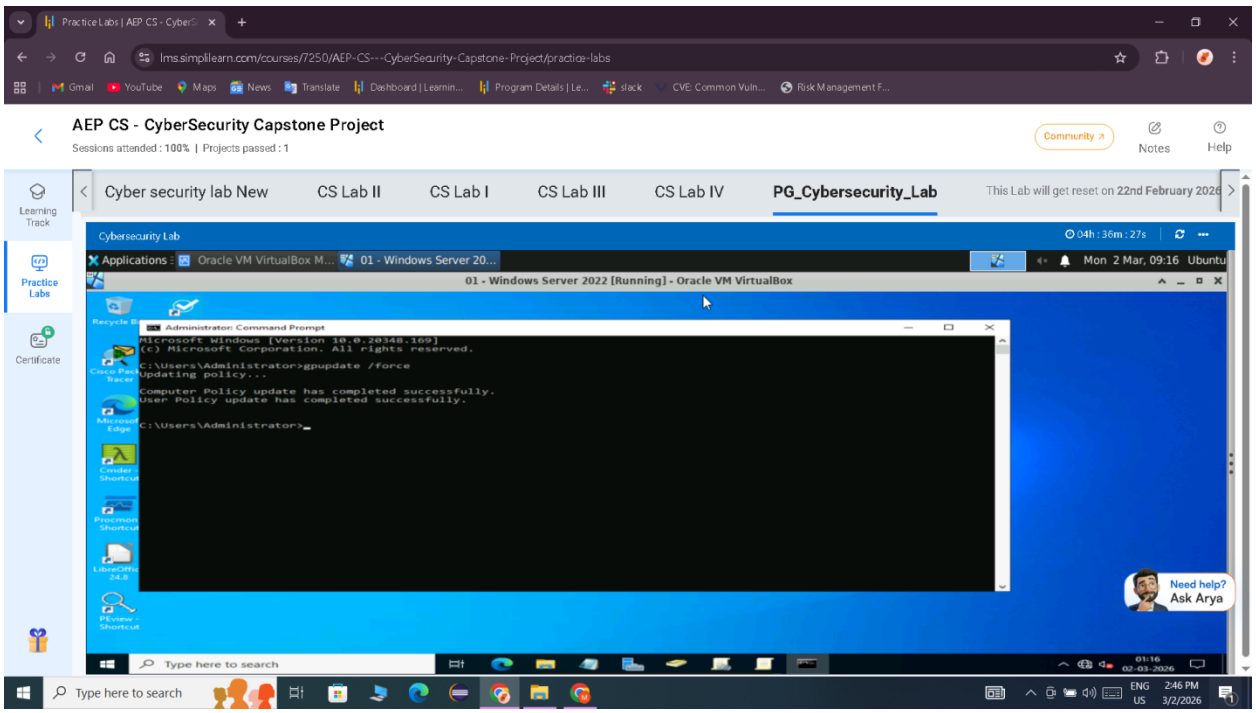
Description: To enforce network security across the enterprise domain, Group Policy Management was utilized. A new Group Policy Object (GPO) named `Enforce_Windows_Firewall` was created and linked directly to the `cyber.local` root domain. The policy was edited to force the Windows Defender Firewall state to "On" for the Domain Profile, ensuring all connected machines comply with baseline security standards.



Group Policy Management Editor confirming the master Domain Profile firewall state is configured to 'On'.

Phase 6: Verifying Security Policy Enforcement

Description: To guarantee the newly created Group Policy Object was actively applied to the domain without waiting for a scheduled refresh cycle, a manual policy update was triggered via the command line interface using the `gpupdate /force` command.



Execution of the gpupdate command, proving the computer and user security policies were successfully pulled and applied.

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6. Final Project Conclusion

This project successfully demonstrated the end-to-end deployment and configuration of a centralized Active Directory environment. By establishing a primary Domain Controller (`DC01`), structuring logical Organizational Units, provisioning secure user accounts, and deploying domain-wide Group Policy Objects to strictly enforce Windows Defender Firewall rules, the network's overall security posture was vastly improved. This centralized infrastructure effectively resolves the challenges of unauthorized access, inefficient user management, and weak policy enforcement highlighted in the initial scenario.